

Titanic200626

June 20, 2026

```
[1]: # Importar las librerías necesarias
import pandas as pd
import matplotlib.pyplot as plt
```

```
[2]: # Cargar mis datos
data = pd.read_csv("2titanic.csv")
```

```
[3]: # Leer los primeros 5 renglones de mi dataset

data.head()
```

```
[3]: PassengerId  Survived  Pclass  \
0              1         0        3
1              2         1        1
2              3         1        3
3              4         1        1
4              5         0        3
```

```

                                Name      Sex  Age  SibSp  \
0                Braund, Mr. Owen Harris   male  22.0     1
1  Cumings, Mrs. John Bradley (Florence Briggs Th... female  38.0     1
2                Heikkinen, Miss. Laina   female  26.0     0
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)   female  35.0     1
4                Allen, Mr. William Henry   male  35.0     0
```

```

Parch      Ticket      Fare Cabin Embarked
0      0    A/5 21171   7.2500   NaN        S
1      0    PC 17599  71.2833   C85        C
2      0  STON/O2. 3101282   7.9250   NaN        S
3      0    113803  53.1000  C123        S
4      0    373450   8.0500   NaN        S
```

```
[4]: # Para ver la forma de mi dataset es decir renglones y columnas
data.shape
```

```
[4]: (891, 12)
```

```
[5]: # Cuántos renglones tiene cada columna
data.count()
```

```
[5]: PassengerId    891
      Survived      891
      Pclass       891
      Name         891
      Sex          891
      Age         714
      SibSp        891
      Parch        891
      Ticket       891
      Fare         891
      Cabin        204
      Embarked     889
      dtype: int64
```

```
[6]: col_names = data.columns.tolist() #Convierta los nombres de las columnas en una
      ↪ lista y las guarde en la variable col_names
col_names
```

```
[6]: ['PassengerId',
      'Survived',
      'Pclass',
      'Name',
      'Sex',
      'Age',
      'SibSp',
      'Parch',
      'Ticket',
      'Fare',
      'Cabin',
      'Embarked']
```

```
[7]: for column in col_names:
      print("Valores nulos en", column, ":", data[column].isnull().sum()) #Esta
      ↪ funcion hace una sumatoria de todos los valores nulos de cada columna
```

```
Valores nulos en PassengerId : 0
Valores nulos en Survived : 0
Valores nulos en Pclass : 0
Valores nulos en Name : 0
Valores nulos en Sex : 0
Valores nulos en Age : 177
Valores nulos en SibSp : 0
Valores nulos en Parch : 0
Valores nulos en Ticket : 0
Valores nulos en Fare : 0
```

Valores nulos en Cabin : 687

Valores nulos en Embarked : 2

```
[8]: # Vamos a revisar la estadística descriptiva de mi dataset
data.describe()
```

```
[8]:
```

	PassengerId	Survived	Pclass	Age	SibSp \
count	891.000000	891.000000	891.000000	714.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008
std	257.353842	0.486592	0.836071	14.526497	1.102743
min	1.000000	0.000000	1.000000	0.420000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000

	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

```
[9]: data.head()
```

```
[9]:
```

	PassengerId	Survived	Pclass \
0	1	0	3
1	2	1	1
2	3	1	3
3	4	1	1
4	5	0	3

	Name	Sex	Age	SibSp \
0	Braund, Mr. Owen Harris	male	22.0	1
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1
2	Heikkinen, Miss. Laina	female	26.0	0
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1
4	Allen, Mr. William Henry	male	35.0	0

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	S
1	0	PC 17599	71.2833	C85	C
2	0	STON/O2. 3101282	7.9250	NaN	S
3	0	113803	53.1000	C123	S
4	0	373450	8.0500	NaN	S

```
[11]: def sex_transform(x):
        d = {'male':'M','female':'F'} # Esta funcion va a utilizar este diccionario
        ↪ que al recibir un valor que va a ser la clave y va a retornar el valor
        ↪ correspondient
        return(d[x])

data['Sex'] = data['Sex'].apply(sex_transform)
```

```
[12]: data.head()
```

```
[12]: PassengerId  Survived  Pclass  \
0             1         0         3
1             2         1         1
2             3         1         3
3             4         1         1
4             5         0         3

                                     Name Sex   Age  SibSp  Parch  \
0                               Braund, Mr. Owen Harris    M  22.0      1      0
1  Cumings, Mrs. John Bradley (Florence Briggs Th...    F  38.0      1      0
2                               Heikkinen, Miss. Laina    F  26.0      0      0
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)    F  35.0      1      0
4                               Allen, Mr. William Henry    M  35.0      0      0

          Ticket    Fare Cabin Embarked
0      A/5 21171   7.2500   NaN         S
1      PC 17599  71.2833   C85         C
2  STON/O2. 3101282   7.9250   NaN         S
3      113803  53.1000  C123         S
4      373450   8.0500   NaN         S
```

```
[13]: # Copiar la columna sex a una columna nueva que se llame Gender

data['Gender'] = data['Sex']
data.head()
```

```
[13]: PassengerId  Survived  Pclass  \
0             1         0         3
1             2         1         1
2             3         1         3
3             4         1         1
4             5         0         3

                                     Name Sex   Age  SibSp  Parch  \
0                               Braund, Mr. Owen Harris    M  22.0      1      0
1  Cumings, Mrs. John Bradley (Florence Briggs Th...    F  38.0      1      0
2                               Heikkinen, Miss. Laina    F  26.0      0      0
```

3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	F	35.0	1	0
4	Allen, Mr. William Henry	M	35.0	0	0

	Ticket	Fare	Cabin	Embarked	Gender
0	A/5 21171	7.2500	NaN	S	M
1	PC 17599	71.2833	C85	C	F
2	STON/O2. 3101282	7.9250	NaN	S	F
3	113803	53.1000	C123	S	F
4	373450	8.0500	NaN	S	M

[14]: # Cuántos y cuáles pasajeros se subieron sin pagar?

```
data[data.Fare == 0]
```

[14]:	PassengerId	Survived	Pclass	Name	Sex	\
179	180	0	3	Leonard, Mr. Lionel	M	
263	264	0	1	Harrison, Mr. William	M	
271	272	1	3	Tornquist, Mr. William Henry	M	
277	278	0	2	Parkes, Mr. Francis "Frank"	M	
302	303	0	3	Johnson, Mr. William Cahoon Jr	M	
413	414	0	2	Cunningham, Mr. Alfred Fleming	M	
466	467	0	2	Campbell, Mr. William	M	
481	482	0	2	Frost, Mr. Anthony Wood "Archie"	M	
597	598	0	3	Johnson, Mr. Alfred	M	
633	634	0	1	Parr, Mr. William Henry Marsh	M	
674	675	0	2	Watson, Mr. Ennis Hastings	M	
732	733	0	2	Knight, Mr. Robert J	M	
806	807	0	1	Andrews, Mr. Thomas Jr	M	
815	816	0	1	Fry, Mr. Richard	M	
822	823	0	1	Reuchlin, Jonkheer. John George	M	

	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	Gender
179	36.0	0	0	LINE	0.0	NaN	S	M
263	40.0	0	0	112059	0.0	B94	S	M
271	25.0	0	0	LINE	0.0	NaN	S	M
277	NaN	0	0	239853	0.0	NaN	S	M
302	19.0	0	0	LINE	0.0	NaN	S	M
413	NaN	0	0	239853	0.0	NaN	S	M
466	NaN	0	0	239853	0.0	NaN	S	M
481	NaN	0	0	239854	0.0	NaN	S	M
597	49.0	0	0	LINE	0.0	NaN	S	M
633	NaN	0	0	112052	0.0	NaN	S	M
674	NaN	0	0	239856	0.0	NaN	S	M
732	NaN	0	0	239855	0.0	NaN	S	M
806	39.0	0	0	112050	0.0	A36	S	M
815	NaN	0	0	112058	0.0	B102	S	M
822	38.0	0	0	19972	0.0	NaN	S	M

```
[15]: # Tabla cruzada entre supervivencia y sexo
pd.crosstab(data.Survived, data.Sex)
```

```
[15]: Sex          F      M
Survived
0          81  468
1         233  109
```

```
[18]: # Cuantos sobrevivieron en total
data[data.Survived==1].count()
```

```
[18]: PassengerId    342
Survived          342
Pclass            342
Name              342
Sex               342
Age              290
SibSp            342
Parch            342
Ticket           342
Fare             342
Cabin            136
Embarked         340
Gender           342
dtype: int64
```

```
[19]: # Agrupar los datos por clase y sexo, después solo mostrar la suma de los que
      ↪ sobrevivieron
pclass_sex_survival_count = data.groupby(['Pclass', 'Sex'])['Survived'].sum()
pclass_sex_survival_count
```

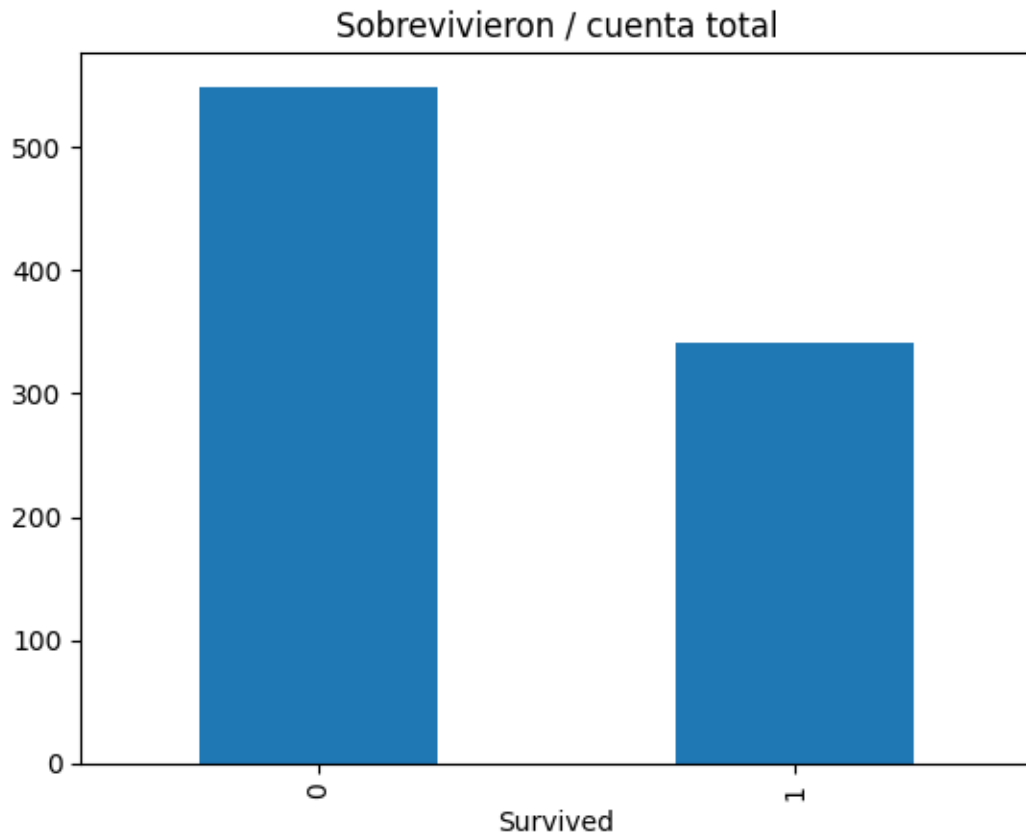
```
[19]: Pclass  Sex
1       F    91
       M    45
2       F    70
       M    17
3       F    72
       M    47
Name: Survived, dtype: int64
```

```
[20]: # Tabla cruzada entre Pclass y Sexo
pd.crosstab(data.Pclass, data.Sex)
```

```
[20]: Sex          F      M
Pclass
1          94  122
2          76  108
3         144  347
```

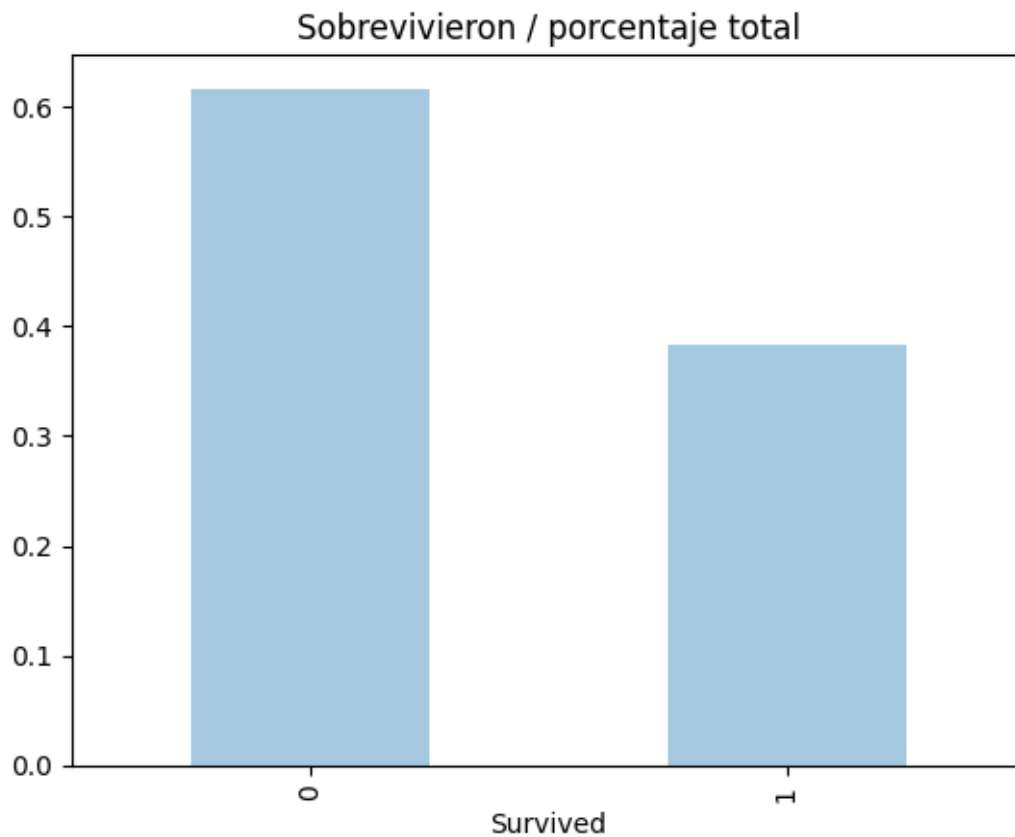
```
[21]: # Gráfica de la columna survived
data.Survived.value_counts().plot(kind = 'bar')
plt.title("Sobrevivieron / cuenta total")
```

```
[21]: Text(0.5, 1.0, 'Sobrevivieron / cuenta total')
```



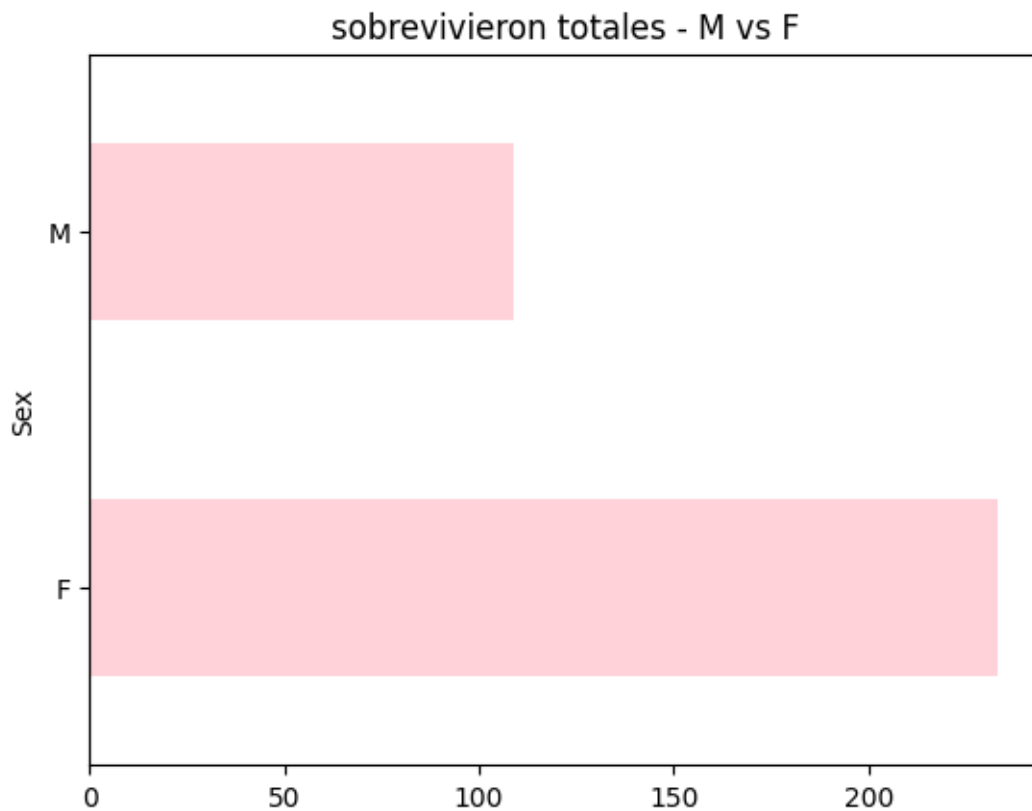
```
[26]: # Gráfica de la columna survived por porcentaje
data.Survived.value_counts(normalize = True).plot(kind = "bar", alpha = 0.4)
plt.title("Sobrevivieron / porcentaje total")
```

```
[26]: Text(0.5, 1.0, 'Sobrevivieron / porcentaje total')
```



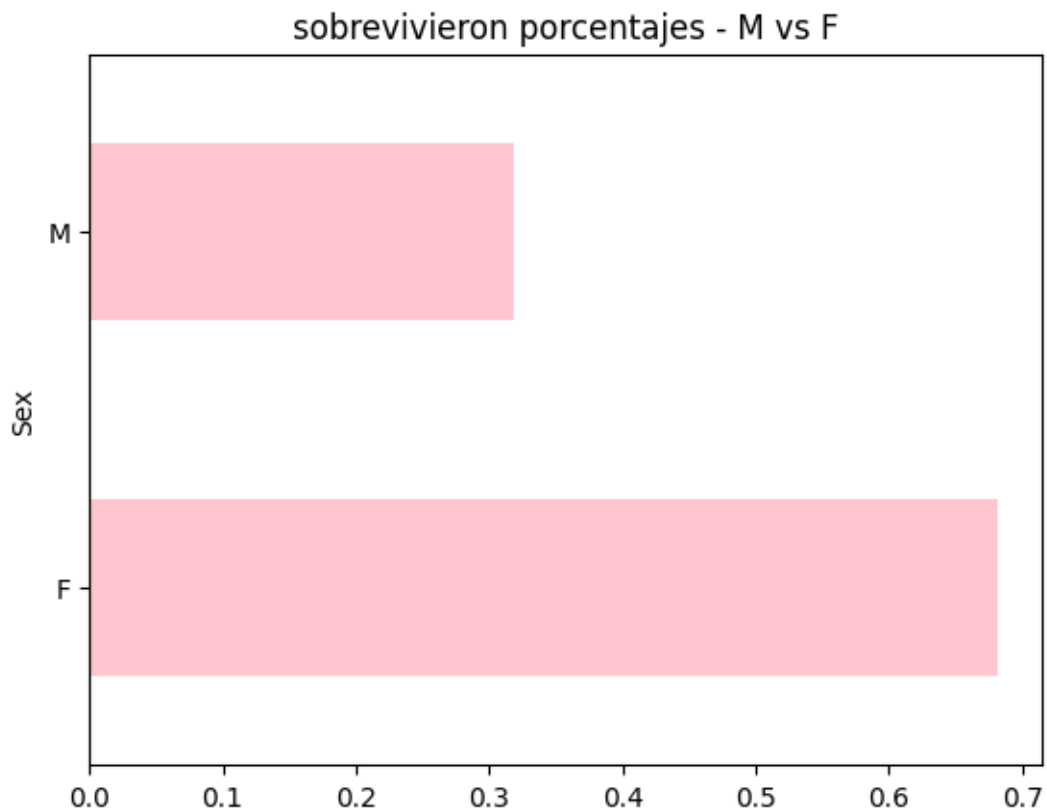
```
[28]: # Gráfica de sexo y survived en cuenta total
data.Sex[data.Survived == 1].value_counts().plot(kind = 'barh', color = "pink",
alpha = 0.7)
plt.title("sobrevivieron totales - M vs F")
```

```
[28]: Text(0.5, 1.0, 'sobrevivieron totales - M vs F')
```

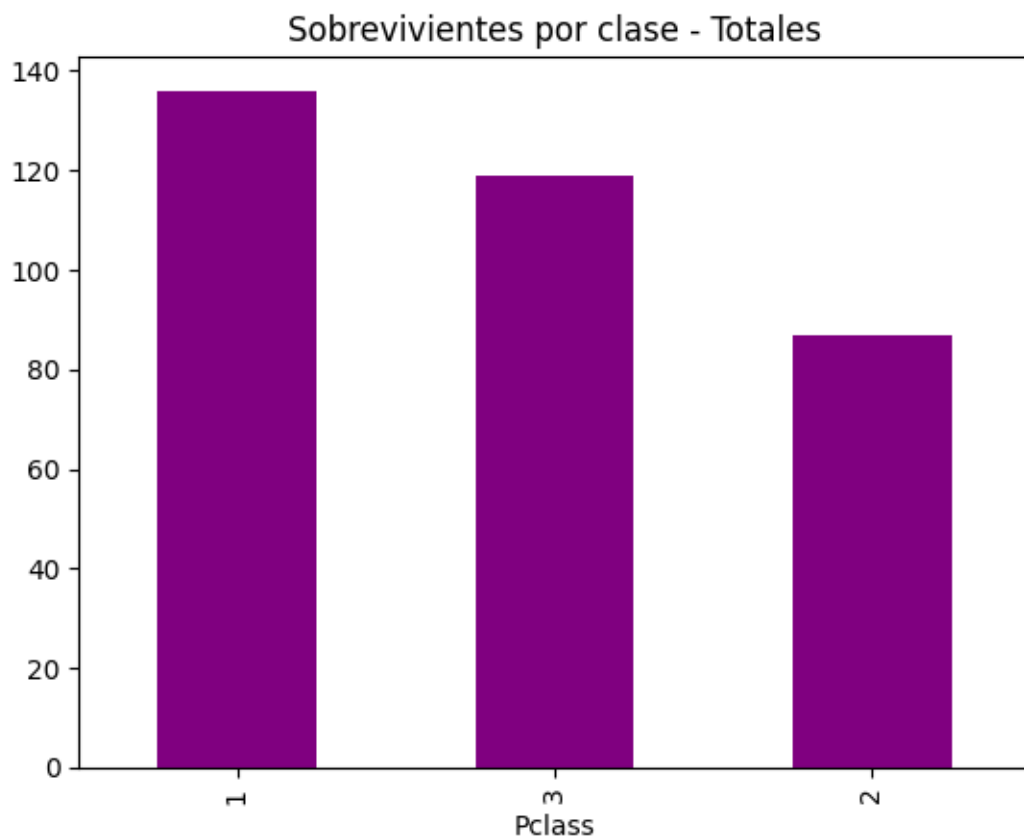
```
[29]: # Gráfica de sexo y survived en porcentajes
data.Sex[data.Survived == 1].value_counts(normalize = True).plot(kind = 'barh',
    color = "pink", alpha = 0.9)
plt.title("sobrevivieron porcentajes - M vs F")
```

```
[29]: Text(0.5, 1.0, 'sobrevivieron porcentajes - M vs F')
```



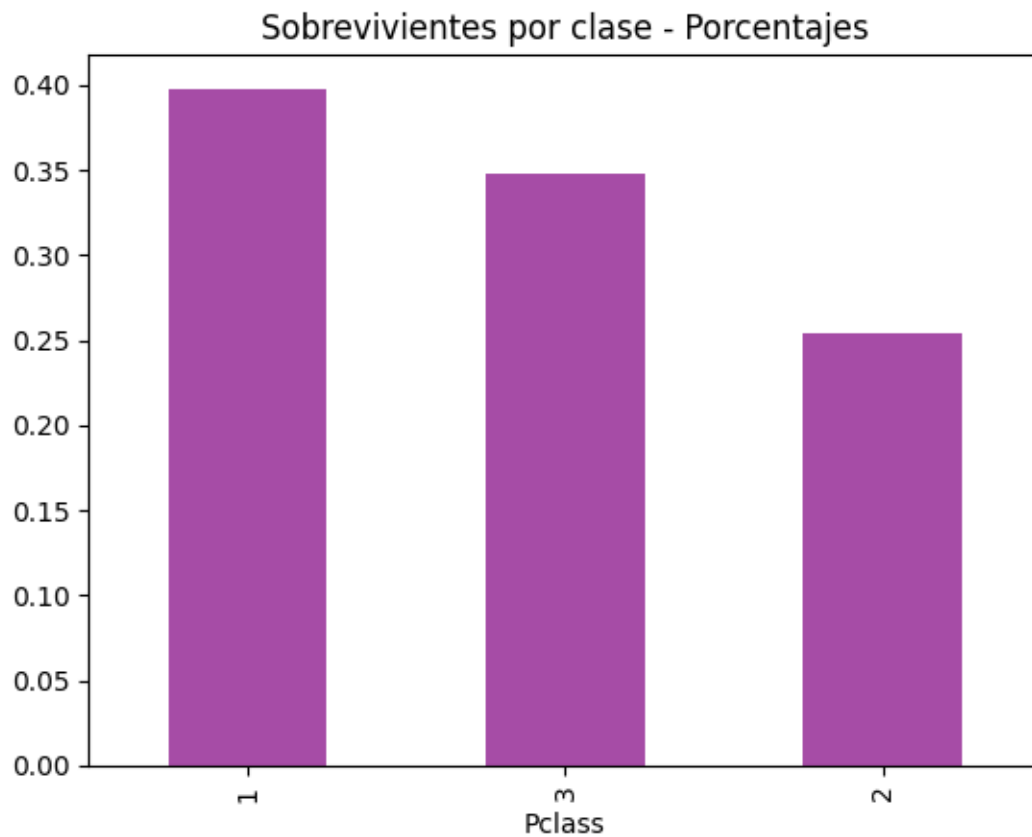
```
[32]: # Gráfica de supervivencia por clase
data.Pclass[data.Survived == 1].value_counts().plot(kind = "bar", color = "purple")
plt.title("Sobrevivientes por clase - Totales")
```

```
[32]: Text(0.5, 1.0, 'Sobrevivientes por clase - Totales')
```



```
[33]: # Gráfica de supervivencia por clase en porcentajes
data.Pclass[data.Survived == 1].value_counts(normalize = True).plot(kind = "
    ↪"bar", color = "purple", alpha = 0.7)
plt.title("Sobrevivientes por clase - Porcentajes")
```

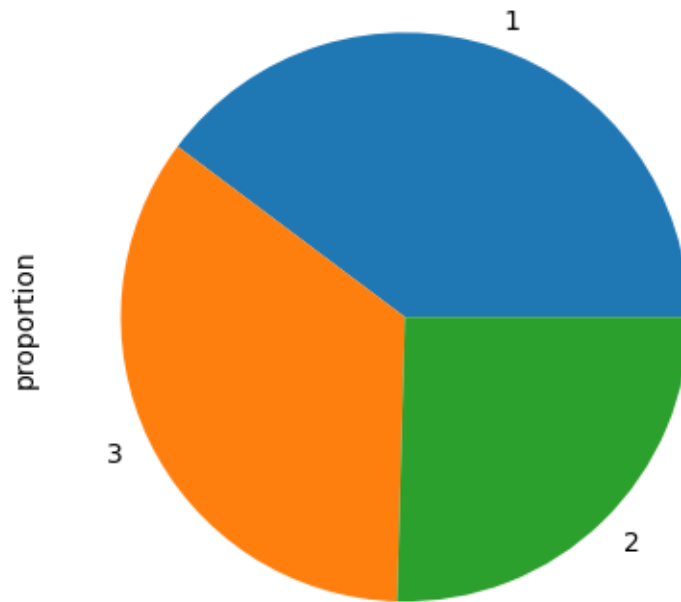
```
[33]: Text(0.5, 1.0, 'Sobrevivientes por clase - Porcentajes')
```



```
[34]: # Gráfica de supervivencia por clase en porcentajes - pastel
data.Pclass[data.Survived == 1].value_counts(normalize = True).plot(kind = "pie")
plt.title("Sobrevivientes por clase - Porcentajes")
```

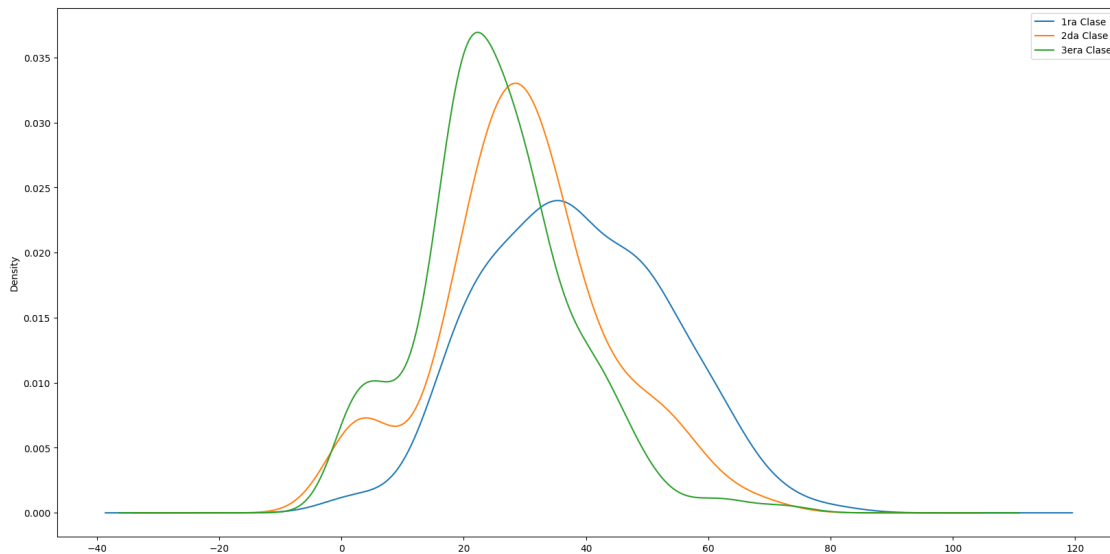
```
[34]: Text(0.5, 1.0, 'Sobrevivientes por clase - Porcentajes')
```

Sobrevivientes por clase - Porcentajes



```
[36]: # Gráfica para ver si hay una relación entre tipo de ticket y edad
fig = plt.figure(figsize = (20,10))
for clase in [1,2,3]:
    data.Age[data.Pclass == clase].plot(kind = "kde")
plt.legend(("1ra Clase", "2da Clase", "3era Clase"))
```

```
[36]: <matplotlib.legend.Legend at 0x316bd6b80>
```



```
[37]: # Información de la persona que pagó más para subirse
data[data.Fare > 510]
```

```
[37]:
```

	PassengerId	Survived	Pclass	Name	Sex	\
258	259	1	1	Ward, Miss. Anna	F	
679	680	1	1	Cardeza, Mr. Thomas Drake Martinez	M	
737	738	1	1	Lesurer, Mr. Gustave J	M	

	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	Gender
258	35.0	0	0	PC 17755	512.3292	NaN	C	F
679	36.0	0	1	PC 17755	512.3292	B51 B53 B55	C	M
737	35.0	0	0	PC 17755	512.3292	B101	C	M

```
[38]: # Cuántos pasajeros eran menores de 1 año
data[data.Age < 1]
```

```
[38]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	\
78	79	1	2	Caldwell, Master. Alden Gates	M	0.83	
305	306	1	1	Allison, Master. Hudson Trevor	M	0.92	
469	470	1	3	Baclini, Miss. Helene Barbara	F	0.75	
644	645	1	3	Baclini, Miss. Eugenie	F	0.75	
755	756	1	2	Hamalainen, Master. Viljo	M	0.67	
803	804	1	3	Thomas, Master. Assad Alexander	M	0.42	
831	832	1	2	Richards, Master. George Sibley	M	0.83	

	SibSp	Parch	Ticket	Fare	Cabin	Embarked	Gender
78	0	2	248738	29.0000	NaN	S	M
305	1	2	113781	151.5500	C22 C26	S	M
469	2	1	2666	19.2583	NaN	C	F

644	2	1	2666	19.2583	NaN	C	F
755	1	1	250649	14.5000	NaN	S	M
803	0	1	2625	8.5167	NaN	C	M
831	1	1	29106	18.7500	NaN	S	M

```
[39]: # Columna de edad en meses
data["Age_months"] = (data["Age"] * 12).round().astype("Int64")
data.head()
```

```
[39]: PassengerId  Survived  Pclass  \
0             1           0         3
1             2           1         1
2             3           1         3
3             4           1         1
4             5           0         3

                                     Name Sex   Age  SibSp  Parch  \
0                               Braund, Mr. Owen Harris    M  22.0      1      0
1  Cumings, Mrs. John Bradley (Florence Briggs Th...    F  38.0      1      0
2                               Heikkinen, Miss. Laina    F  26.0      0      0
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)        F  35.0      1      0
4                               Allen, Mr. William Henry    M  35.0      0      0

      Ticket      Fare Cabin Embarked Gender  Age_months
0      A/5 21171    7.2500   NaN         S      M         264
1      PC 17599   71.2833   C85         C      F         456
2  STON/O2. 3101282    7.9250   NaN         S      F         312
3      113803   53.1000  C123         S      F         420
4      373450    8.0500   NaN         S      M         420
```

```
[40]: data[data.Age < 1]
```

```
[40]: PassengerId  Survived  Pclass                                     Name Sex   Age  \
78             79           1         2  Caldwell, Master. Alden Gates    M  0.83
305            306           1         1  Allison, Master. Hudson Trevor    M  0.92
469            470           1         3  Baclini, Miss. Helene Barbara    F  0.75
644            645           1         3  Baclini, Miss. Eugenie        F  0.75
755            756           1         2  Hamalainen, Master. Viljo    M  0.67
803            804           1         3  Thomas, Master. Assad Alexander    M  0.42
831            832           1         2  Richards, Master. George Sibley    M  0.83

      SibSp  Parch  Ticket      Fare  Cabin Embarked Gender  Age_months
78        0      2  248738   29.0000   NaN         S      M         10
305        1      2  113781  151.5500  C22 C26         S      M         11
469        2      1    2666   19.2583   NaN         C      F          9
644        2      1    2666   19.2583   NaN         C      F          9
755        1      1  250649   14.5000   NaN         S      M          8
```

803	0	1	2625	8.5167	NaN	C	M	5
831	1	1	29106	18.7500	NaN	S	M	10

[]: